Packetized Elementary Stream Headers

Depending on the Stream ID, a Packetized Elementary Stream (PES) header may contain various components.

The header

byte 0	byte 1	byte 2	byte 3	byte 4	byte 5
0000 0000 0000 0000 0000 0001 Start code			Stream ID	PES packet length	

Stream ID's which pertain to DVD

Stream ID	Stream type	extension present?
1011 1101 0xBD	Private stream 1 (non MPEG audio, subpictures)	Yes
1011 1110 0xBE	Padding stream	No
1011 1111 0xBF	Private stream 2 (navigation data)	No
110x xxxx 0xC0 - 0xDF	MPEG-1 or MPEG-2 audio stream number x xxxx note: DVD allows only 8 audio streams	Yes
1110 xxxx 0xE0 - 0xEF	MPEG-1 or MPEG-2 video stream number xxxx note: DVD allows only 1 video stream	Yes

The extension

byte 6								byte 7								byte 8							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
10	PES scrambling control			PES priority	data alignment indicator	copyright	original or copy	PTS DTS flags		ESCR flag	ES rate flag	DSM trick mode flag	additional copy info flag	PES CRC flag	PES extension flag	PES header data length							

The bit fields of the basic extension:

PES scrambling control -- 00 = not scrambled, others are user defined.

PES priority -- provides 2 priority levels, 0 and 1.

data alignment indicator -- if set to 1 indicates that the PES packet header is immediately followed by the video start code or audio syncword.

copyright -- 1 = packet contains copyrighted material.

original or copy -- 1 = original, 0 = copy.

PTS DTS flags -- Presentation Time Stamp / Decode Time Stamp. 00 = no PTS or DTS data present, 01 is forbidden.
if set to 10 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
0010				PTS 32..30			1	PTS 29..15							1	PTS 14..00							1									

PTS DTS flags -- if set to 11 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
0011				PTS 32..30			1	PTS 29..15							1	PTS 14..00							1									
0001				DTS 32..30			1	DTS 29..15							1	DTS 14..00							1									

presentation and decoder time stamps use the 90KHz clock (SCR / 300)

ESCR -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
00	ESCR base 32..30						1	ESCR base 29..15							1	ESCR base 14..00							1	ESCR ext							1							

This is the Elementary Stream Clock Reference, used if the stream and system levels are not synchronized (i.e. ESCR differs from SCR in the PACK header).

ES rate -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
1	ES rate																						1

The rate at which data is delivered for this stream, in units of 50 bytes/second.

DSM trick mode - not used by DVD

additional copy info -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0
1	additional copy info						

PES CRC flag -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
previous PES packet CRC															

The polynomial used is $X^{16} + X^{12} + X^5 + 1$

PES extension flag -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0
PES private data flag	pack header field flag	program packet sequence counter flag	P-STD buffer flag	111			PES extension flag 2

The bit fields of the first extension:

PES private data flag -- if set to 1 16 bytes of user defined data is appended to the header data field.

pack header field flag -- if set to 1 the 8-bit pack field length value is appended to the header data field.

program packet sequence counter flag -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
1	packet sequence counter							1	MPEG1_MPEG2 identifier							original stuffing length							

P-STD buffer flag -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
01	P-STD buffer scale 0 = 128 bytes 1 = 1024 bytes							P-STD buffer size							

PES extension flag 2 -- if set to 1 the following data is appended to the header data field:

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
1	PES extension field length							reserved							

Which is then followed by the indicated number of bytes.

The rest of the packet

Following all the extension data are optional stuffing bytes (0xFF), which are included in the header data length, and then the payload.
In the case of private streams the first byte of the payload is the sub-stream number.

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